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ORIGINAL ARTICLE

A comprehensive analysis of stroke risk factors and development of a predictive model using machine learning approaches

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Abstract

Stroke is a leading cause of death and disability globally, particularly in China. Identifying risk factors for stroke at an early stage is critical to improving patient outcomes and reducing the overall disease burden. However, the complexity of stroke risk factors requires advanced approaches for accurate prediction. The objective of this study is to identify key risk factors for stroke and develop a predictive model using machine learning techniques to enhance early detection and improve clinical decision-making. Data from the China Health and Retirement Longitudinal Study (2011–2020) were analyzed, classifying participants based on baseline characteristics. We evaluated correlations among 12 chronic diseases and applied machine learning algorithms to identify stroke-associated parameters. A dose–response relationship between these parameters and stroke was assessed using restricted cubic splines with Cox proportional hazards models. A refined predictive model, incorporating age, sex, and key risk factors, was developed. Stroke patients were significantly older (average age 69.03 years) and had a higher proportion of women (53%) compared to non-stroke individuals. Additionally, stroke patients were more likely to reside in rural areas, be unmarried, smoke, and suffer from various diseases. While the 12 chronic diseases were correlated ($p < 0.05$), the correlation coefficients were generally weak ($r < 0.5$). Machine learning identified nine parameters significantly associated with stroke risk: TyG-WC, WHtR, TyG-BMI, TyG, TMO, CysC, CREA, SBP, and HDL-C. Of these, TyG-WC, WHtR, TyG-BMI, TyG, CysC, CREA, and SBP exhibited a positive dose–response relationship with stroke risk. In contrast, TMO and HDL-C were associated with reduced stroke risk. In the fully adjusted model, elevated CysC (HR = 2.606, 95% CI 1.869–3.635), CREA (HR = 1.819, 95% CI 1.240–2.668), and SBP (HR = 1.008, 95% CI 1.003–1.012) were significantly associated with increased stroke risk, while higher HDL-C (HR = 0.989, 95% CI 0.984–0.995) and TMO (HR = 0.99995, 95% CI 0.99994–0.99997) were protective. A nomogram model incorporating age, sex, and the identified parameters demonstrated superior predictive accuracy, with a significantly higher Harrell's C-index compared to individual predictors. This study identifies several significant stroke risk factors and presents a predictive model that can enhance early detection of high-risk individuals. Among them, CREA, CysC, SBP, TyG-BMI, TyG, TyG-WC, and WHtR were positively associated with stroke risk, whereas TMO and HDL-C were opposite. This serves as a valuable decision-support resource for clinicians, facilitating more effective prevention and treatment strategies, ultimately improving patient outcomes.

Keywords Stroke · Risk factors · Machine learning · Feature parameters · Nomogram predictive modelling · Dose–response relationship

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Introduction

Stroke is widely recognized as the second leading cause of death and the second most significant cause of severe disability worldwide (Timmis et al. 2021; Thayabaranathan et al. 2022). Stroke refers to the damage caused to cerebral blood vessels by various factors, resulting in localized (or widespread) brain tissue injury that leads to clinical symptoms persisting for more than 24 h or causing death (Bersano and Gatti 2023). It is characterized by high rates of incidence, disability, recurrence, and mortality. According to a 2019 statistical survey, stroke claimed 6.55 million lives and resulted in a loss of 143 million disability-adjusted life years (DALYs) (Global, regional, and national burden of stroke and its risk factors 1990–2019: a systematic analysis for the Global Burden of Disease Study 2019 2019). In China alone, over 2 million new cases of stroke occured annually (Fan et al. 2023a). In 2020, the hospitalization costs related to stroke in China amounted to 58 billion CNY, with patients themselves covering approximately 19.8 billion CNY (Tu and Wang 2021). Beyond the direct medical expenses, rehabilitation costs and other disease-related losses have placed a significant financial burden on both society and individuals (Feigin and Owolabi 2023). Due to the coexistence of cardiovascular and cerebrovascular conditions, the prognosis for stroke patients is often poor. Preventive measures and effective treatment can improve outcomes and survival rates (Benjamin et al. 2018; Stinear et al. 2020).Therefore, practical solutions are urgently needed to alleviate the burden of stroke and its related complications, ultimately saving lives and improving global health, quality of life, and socioeconomic productivity.

Stroke can be classified into ischemic stroke, caused by vascular obstruction, and hemorrhagic stroke, resulting from vascular rupture (Diener and Hankey 2020). Approximately 84.4% of strokes are ischemic (Yang et al. 2023). Identifying risk factors for stroke is crucial for recognizing high-risk populations and ensuring early intervention to reduce morbidity and mortality. Clinically, stroke risk factors are numerous. The first category includes non-modifiable risk factors such as age, gender, ethnicity, and genetics (Tu and Wang 2021). The second category encompasses modifiable risk factors, including diseases related to stroke such as hypertension, diabetes, dyslipidemia, and heart disease, as well as lifestyle factors such as smoking, alcohol consumption, diet, obesity, physical inactivity, and psychological stress (Tu and Wang 2021; Guzik and Bushnell 2017). Additionally, factors, including education level (Fan et al. 2023a), economic status (Potter et al. 2022), and environmental factors (Bhatnagar 2022), are often considered relevant to stroke occurrence. Several specific biomarkers are associated with stroke risk. Researchers have developed indicators like the Triglyceride-glucose index (TyG) and its modified version, which may not only be influencing factors for stroke but can also be used to assess stroke risk (Yang et al. 2023; Che et al. 2023; Yu et al. 2024). Other predictive markers, such as body mass index (BMI) (Marini et al. 2020), high-density lipoprotein cholesterol (HDL-C) (Fan et al. 2023b) and cystatin C (CysC) (Liu et al. 2021), are also associated with stroke risk.

However, relying solely on conventional risk factors is insufficient to comprehensively explain all potential risks associated with stroke, thus limiting the predictive accuracy of these factors (Wang et al. 2022; Shao et al. 2024). There is a pressing need in clinical practice to investigate further modifiable risk factors that may contribute to stroke development. Given the complexity and diversity of factors leading to stroke, advanced machine learning technologies have emerged as valuable tools in stroke research (Chavva et al. 2022). Machine learning algorithms can seamlessly integrate numerous variables (Stahlschmidt et al. 2022), enabling comprehensive analysis of vast datasets, including demographic and clinical factors, to better predict stroke risk. In this study, we utilized statistical data from the China Health and Retirement Longitudinal Study (CHARLS) from 2011 to 2020 to conduct a comprehensive analysis of stroke risk factors and develop predictive models using various machine learning methods.

Materials and methods

Data acquisition and preprocessing

Data were obtained from the China Health and Retirement Longitudinal Study (CHARLS, http://charls.pku.edu.cn/), covering statistical data from 2011 to 2020, with approval from the Institutional Review Board of Peking University (IRB00001052-11015). To ensure data quality and minimize bias, a systematic preprocessing workflow was applied. First, samples with missing stroke information and those with over 5% missing data were excluded, along with patients diagnosed with cancer. Missing data patterns for the top 20 variables were visualized using the VIM package in R to identify potential issues (Kowarik and Templ 2016). Remaining missing values were handled through multiple imputation using the MICE package, where variables were categorized into binary, unordered categorical, ordered categorical, or continuous types, and appropriate imputation methods (e.g., logistic regression, polytomous regression, proportional odds model, and predictive mean matching) were applied (Aarts 2010). The imputation process involved 5 iterations with 50 imputations each, efficiently executed

using 4 CPU cores. Following imputation, the datasets were combined into a final complete dataset, which was validated to ensure no remaining missing values. Continuous variables were then screened for outliers using the 3-sigma rule, and any identified outliers were removed (Leys et al. 2013). To assess the robustness of the imputation process, sensitivity analyses were conducted by comparing model performance on the original dataset with missing values and the imputed dataset, confirming no significant bias was introduced. The resulting clean dataset comprised 12,978 patients and was used for subsequent analysis.

To address class imbalance in the dataset, where stroke cases accounted for only 1011 out of 12,978 participants, several strategies were implemented. Resampling techniques, including oversampling the minority class using the Synthetic Minority Over-sampling Technique (SMOTE) and undersampling the majority class, were applied. For algorithms supporting class weights (e.g., Random Forest, SVM), weights inversely proportional to class frequencies were incorporated to penalize misclassification of the minority class. Robust ensemble methods such as Random Forest and Gradient Boosting were employed, and classification thresholds were optimized to balance sensitivity and specificity. Evaluation metrics less sensitive to class imbalance, including the area under the receiver operating characteristic curve (AUROC) and the F1-score, were used alongside accuracy, and stratified k-fold cross-validation ensured consistent class proportions across folds. Following these steps, the final dataset comprising 12,978 participants was prepared for subsequent analysis and predictive modeling.

Baseline characteristics of the study population

Baseline characteristics were analyzed using the gtsummary package in R. Variables were categorized as either continuous or categorical based on their data type and the number of unique values. A summary table was generated, stratifying the data by stroke status. For categorical variables, frequencies and percentages were reported. For continuous variables, means and standard deviations were calculated. Overall statistics were included, and p-values were computed to assess differences between groups.

Disease correlation analysis

Correlations between 12 chronic diseases were analyzed using R. The diseases included hypertension, diabetes, pulmonary disease, dyslipidemia, heart disease, stroke, psyche, liver disease, kidney disease, digestive disease, asthma, and memory disorder. Phi coefficients were calculated to measure the association between these binary disease variables. A correlation matrix was generated, and p-values were computed to assess the statistical significance of the correlations. The corrplot package was used to visualize the results as a bubble plot, with the lower triangle showing significance levels and the upper triangle displaying correlation coefficients. The plot used a color gradient to represent correlation strength, with significance levels indicated by asterisks.

Machine learning-based feature selection and importance analysis

To comprehensively identify and rank important features associated with stroke risk, we employed eight machine learning algorithms: Elastic Net, Lasso, Gradient Boosting Machine (GBM), Random Forest, XGBoost, Neural Network, Support Vector Machine (SVM), and Logistic Regression. The selection of these algorithms was based on their well-established performance in medical prediction tasks and their ability to handle diverse data characteristics. Elastic Net and Lasso were chosen for their effectiveness in feature selection and mitigating multicollinearity, while Random Forest, GBM, and XGBoost excel at capturing nonlinear relationships and interactions. Neural Network methods were included for their capacity to model complex patterns in high-dimensional data, and SVM was selected for its robustness in binary classification tasks. Logistic Regression served as a baseline model, offering interpretable results to ensure clinical relevance. To ensure unbiased performance estimation, we implemented fivefold cross-validation for all algorithms and evaluated their predictive power using ROC-AUC, sensitivity, and specificity metrics. Feature importance scores were calculated for each algorithm, normalized, and visualized using bar plots to highlight the top 10 features identified by each method. To assess the robustness and consensus among the different algorithms, a scatter plot was generated to display the average importance score against the number of occurrences for each feature across all methods. Features consistently ranked as important were prioritized, with particular emphasis on the top five features by occurrence count and average importance score. By integrating diverse machine learning approaches, our methodology ensured a robust and thorough exploration of the dataset, capturing both linear and non-linear relationships, minimizing algorithm-specific bias, and identifying the most critical predictors of stroke risk with high confidence.

Dose–response relationship analysis using restricted cubic splines

The dose–response relationships between selected predictors and stroke were analyzed using restricted cubic splines (RCS) in Cox proportional hazards models. The top features identified by machine learning algorithms were used as predictors. Cox regression models were fitted, adjusting for age, gender, residence, education, marital status, smoking, and

alcohol consumption. RCS with three knots were applied to capture potential non-linear relationships. Hazard ratios and 95% confidence intervals were calculated, with the median value of each predictor set as the reference point. The density distribution of each predictor was overlaid on the hazard ratio plot to provide context for the observed relationships.

Cox proportional hazards regression analysis

Cox proportional hazards regression models were employed to assess the association between selected predictors and stroke risk. Three models with increasing levels of adjustment were constructed: Model 1 adjusted for age and gender; Model 2 additionally adjusted for marital status, residence, education, smoking status, and alcohol consumption; Model 3 further adjusted for hypertension, diabetes, heart disease, dyslipidemia, kidney disease, systolic blood pressure, and diastolic blood pressure. Hazard ratios (HRs), 95% confidence intervals (CIs), and p-values were calculated for each predictor across all three models.

Predictive model development and evaluation

A Cox proportional hazards model was developed to predict stroke risk, incorporating selected variables identified by machine learning algorithms, as well as age and gender. The model was visualized using a nomogram to facilitate individualized risk estimation. Calibration curves were generated for 1-, 3-, and 5-year predictions using bootstrap resampling with 500 iterations, allowing assessment of the model's accuracy by comparing predicted probabilities with observed event rates. Time-dependent area under the curve (AUC) analysis was performed to evaluate the model's discriminative ability over an 8-year follow-up period. Additionally, Harrell's C-index was calculated to quantify the model's overall discrimination performance, with values above 0.70 indicating good predictive power.

Results

Characteristics of the study population and their disease associations

A total of 12,978 participants were included in this study, of whom 1011 had experienced a stroke, as shown in Table 1. In terms of demographic characteristics, stroke patients were generally older than non-stroke participants, and a higher proportion of stroke patients were rural residents, women, and smokers ($p < 0.001$). Additionally, a higher percentage of women and individuals in married status were observed in the stroke group ($p < 0.001$). Furthermore, compared to non-stroke participants, stroke patients exhibited significantly higher values for weight, waist circumference, BMI, TyG index, and TyG-BMI index ($p < 0.001$). Regarding comorbidities, the prevalence of hypertension, diabetes, dyslipidemia, heart disease, arthritis, lung disease, liver disease, kidney disease, asthma, mental illness, and memory impairment were notably higher in the stroke group compared to the non-stroke group ($p < 0.001$). There was also a significant increase in the prevalence of digestive system diseases ($p < 0.05$). Blood tests and metabolic analyses revealed that stroke patients had significantly elevated levels of glucose, creatinine (CREA), and triglycerides (TG) ($p < 0.05$), whereas levels of HDL-C and total metabolic output were significantly lower ($p < 0.001$). No significant differences were found in the levels of mean corpuscular volume, blood platelet count, or blood urea nitrogen ($p > 0.05$). These changes may reflect metabolic disorders in stroke patients. In addition, depression scores were significantly higher in stroke patients ($p < 0.001$), which may be related to the psychological impact following a stroke. We further analyzed the associations between 12 different diseases. As shown in Fig. 1, although stroke was significantly correlated with other diseases ($p < 0.05$), the correlation coefficients were generally low. This suggests that stroke is a multifactorial disease likely involving complex pathophysiological mechanisms.

Identification of stroke-related parameters based on machine learning

To effectively evaluate stroke risk, we employed eight advanced machine learning algorithms to analyze the features associated with stroke. Figure 2A illustrates the top 10 ranked parameters across each algorithm. The eight algorithms—ElasticNet, GBM, Lasso, Logistic Regression, Neural Network, Random Forest, SVM, and XGBoost—prioritized different parameters: waist-to-height ratio (WHtR), systolic blood pressure (SBP), CysC, WHtR, TyG, TyG-BMI, CREA, and SBP. To further assess the significance of these parameters, we evaluated both the importance scores and the frequency of their occurrence among the top five parameters for each algorithm. As shown in Fig. 2B, we identified nine key feature parameters. The top five ranked parameters based on total importance score were TyG-waist-circumference (WC), WHtR, TyG-BMI, TyG, and total metabolic output (TMO), with TyG-WC achieving the highest score of 0.74. Additionally, CysC appeared most frequently across different algorithms, appearing seven times, which underscores its importance across multiple models. Other frequently occurring parameters included WHtR, CREA, SBP, and HDL-C, which also appeared in several algorithms with high frequencies. Ultimately, we identified nine critical feature parameters: TyG-WC, WHtR, TyG-BMI, TyG, TMO, CysC, CREA, SBP, and HDL-C.

Table 1 Baseline characteristics of study participants stratified by stroke status

Characteristic	Overall N = 12,978	Non-stroke N = 11,967	Stroke N = 1011	p-value
Age, years, mean (SD)	64.80 (9.93)	64.45 (9.93)	69.03 (8.91)	< 0.001
Gender, n (%)				< 0.001
Male, n (%)	7075 (54.52%)	6600 (55.15%)	475 (46.98%)	
Female, n (%)	5903 (45.48%)	5367 (44.85%)	536 (53.02%)	
Residence, n (%)				0.6
Urban	4970 (38.29%)	4590 (38.36%)	380 (37.59%)	
Rural	8008 (61.70%)	7377 (61.64%)	631 (62.41%)	
Marry, n (%)				< 0.001
No	2347 (18.08%)	2111 (17.64%)	236 (23.34%)	
Yes	10,631 (81.92%)	9856 (82.36%)	775 (76.66%)	
Smoking, n (%)				< 0.001
No	7554 (58.21%)	7044 (58.86%)	510 (50.61%)	
Yes	5424 (41.79%)	4923 (41.14%)	501 (49.55%)	
Alcohol-drinking, n (%)				0.5
No	7079 (54.55%)	6537 (54.63%)	542 (53.61%)	
Yes	5899 (45.45%)	5430 (45.47%)	469 (46.39%)	
Height, m, mean (SD)	1.58 (0.08)	1.58 (0.08)	1.58 (0.09)	> 0.9
Weight, kg, mean (SD)	59.50 (10.95)	59.38 (10.91)	60.93 (11.36)	< 0.001
Waist, cm, mean (SD)	86.02 (10.15)	85.82 (10.09)	88.38 (10.62)	< 0.001
BMI, kg/m^2, mean (SD)	23.75 (3.58)	23.70 (3.58)	24.33 (3.55)	< 0.001
Hypertension, n (%)				< 0.001
No	7644 (58.90%)	7352 (61.44%)	292 (28.88%)	
Yes	5334 (41.10%)	4615 (38.56%)	719 (71.12%)	
Diabetes, n (%)				< 0.001
No	11,361 (87.54%)	10,590 (88.49%)	771 (76.26%)	
Yes	1617 (12.50%)	1377 (11.51%)	240 (23.74%)	
Dyslipidemia, n (%)				< 0.001
No	9659 (74.43%)	9126 (76.26%)	533 (52.72%)	
Yes	3319 (25.57%)	2841 (23.74%)	478 (47.28%)	
Heart-disease, n (%)				< 0.001
No	9995 (77.01%)	9389 (78.46%)	606 (59.94%)	
Yes	2983 (22.98%)	2578 (21.54%)	405 (40.06%)	
Arthritis, n (%)				< 0.001
No	7760 (59.79%)	7266 (60.72%)	494 (48.86%)	
Yes	5218 (40.21%)	4701 (39.28%)	517 (51.14%)	
Pulmonary-disease, n (%)				< 0.001
No	10,901 (84.00%)	10,106 (84.45%)	795 (78.64%)	
Yes	2,077 (16.00%)	1,861 (15.55%)	216 (21.36%)	
Liver-disease, n (%)				< 0.001
No	12,065 (92.97%)	11,168 (93.32%)	897 (88.72%)	
Yes	913 (7.03%)	799 (6.68%)	114 (11.28%)	
Kidney-disease, n (%)				< 0.001
No	11,610 (89.46%)	10,789 (90.16%)	821 (81.21%)	
Yes	1368 (10.54%)	1178 (9.84%)	190 (18.79%)	
Digestive-disease, n (%)				0.011
No	8,797 (67.78%)	8,148 (68.09%)	649 (64.19%)	
Yes	4,181 (32.22%)	3,819 (31.91%)	362 (35.81%)	
Asthma, n (%)				< 0.001
No	12,047 (92.83%)	11,136 (93.06%)	911 (90.11%)	

Table 1 (continued)

Characteristic	Overall N = 12,978	Non-stroke N = 11,967	Stroke N = 1011	p-value
Yes	931 (7.17%)	831 (6.94%)	100 (9.89%)	
Psyche-disease, n (%)				< 0.001
No	12,582 (96.95%)	11,654 (97.38%)	928 (91.79%)	
Yes	396 (3.05%)	313 (2.62%)	83 (8.21%)	
Memory-disorder, n (%)				< 0.001
No	12,263 (94.49%)	11,458 (95.7%)	805 (79.62%)	
Yes	715 (5.51%)	509 (4.25%)	206 (20.38%)	
White-blood-cell, /L, mean (SD)	5.93 (1.60)	5.91 (1.60)	6.14 (1.62)	< 0.001
Mean-corpuscular-volume, fL, mean (SD)	91.71 (6.70)	91.71 (6.71)	91.66 (6.61)	0.5
Blood-platelet-count, /L, mean (SD)	204.52 (64.08)	204.47 (64.01)	205.14 (64.93)	0.8
Blood-urea-nitrogen, mmol/L, mean (SD)	15.27 (4.16)	15.26 (4.16)	15.34 (4.24)	0.8
Glucose, mg/dL, mean (SD)	99.70 (19.39)	99.42 (19.04)	103.04 (22.90)	< 0.001
CREA, mg/dL, mean (SD)	0.78 (0.18)	0.78 (0.17)	0.82 (0.20)	< 0.001
TC, mg/dL, mean (SD)	183.59 (33.63)	183.44 (33.53)	185.29 (34.77)	0.064
TG, mg/dL, mean (SD)	129.68 (68.97)	128.73 (68.44)	141.01 (74.01)	< 0.001
HDL-C, mg/dL, mean (SD)	51.22 (11.23)	51.40 (11.21)	49.14 (11.26)	< 0.001
LDL-C, mg/dL, mean (SD)	105.07 (28.71)	104.93 (28.64)	106.67 (29.40)	0.035
CRP, mg/dL, mean (SD)	2.00 (2.51)	1.97 (2.49)	2.34 (2.69)	< 0.001
Glycosylated-hemoglobin, mmol/L, mean (SD)	5.70 (0.62)	5.69 (0.61)	5.82 (0.71)	< 0.001
Uric-acid, mg/dL, mean (SD)	4.77 (1.28)	4.75 (1.27)	5.03 (1.39)	< 0.001
Hematocrit-value, %, mean (SD)	41.43 (4.96)	41.40 (4.94)	41.86 (5.12)	0.007
Hemoglobin, g/dL, mean (SD)	13.80 (1.72)	13.78 (1.72)	14.01 (1.75)	< 0.001
CysC, mg/L, mean (SD)	0.86 (0.20)	0.86 (0.19)	0.94 (0.21)	< 0.001
TMO, kcal/g/h, mean (SD)	5505.88 (5529.88)	5653.45 (5565.29)	3759.10 (4758.48)	< 0.001
TyG index, mean (SD)	8.64 (0.55)	8.63 (0.55)	8.75 (0.58)	< 0.001
TyG-BMI, mean (SD)	205.40 (37.75)	204.77 (37.68)	212.85 (37.81)	< 0.001
SBP, mmHg, mean (SD)	127.01 (19.34)	126.23 (18.99)	136.27 (20.98)	< 0.001
DBP, mmHg, mean (SD)	74.80 (11.18)	74.51 (11.07)	78.28 (11.89)	< 0.001
CESD-10, scores, mean (SD)	8.84 (6.39)	8.62 (6.30)	11.40 (6.89)	< 0.001

CREA creatinine, *TC* total-cholesterol, *TG* triglycerides, *HDL-C* high-density-lipoprotein-cholesterol, *LDL-C* low-density-lipoprotein-cholesterol, *CRP* C-reaction-protein, *CysC* cystatin-C, *TMO* total metabolic output, *TyG* triglyceride-glucose-index, *BMI* body-mass-index, *SBP* systolic blood pressure, *DBP* diastolic blood pressure

Dose–response relationship and risk association of key predictive factors for stroke

We used the nine feature parameters identified through the machine learning algorithms (TyG-WC, WHtR, TyG-BMI, TyG, TMO, CysC, CREA, SBP, and HDL-C) as predictive factors and analyzed their dose–response relationships with stroke. As shown in Fig. 3, CREA, CysC, SBP, TyG-BMI, TyG, TyG-WC, and WHtR exhibited significant positive dose–response relationships with stroke risk. This indicates that as the levels of these predictive factors increase, the risk of stroke also rises. The relationship followed an exponential curve, suggesting that when these parameters reach a certain threshold, stroke risk may enter a "high-risk acceleration zone," where the risk increased substantially. Conversely, TMO and HDL-C demonstrated a negative dose-dependent pattern with stroke risk. This implies that a decrease in TMO levels and an increase in HDL-C levels are associated with a reduction in stroke risk. These findings underscore the importance of monitoring and managing these key parameters in stroke prevention strategies.

Table 2 presents the specific associations between the nine predictive factors and stroke risk, as analyzed using the Cox regression model. Across the three progressively adjusted models, we observed changes in the correlations between these predictive factors and stroke risk as the models were refined. In the fully adjusted Model 3, elevated levels of CysC, CREA, and SBP were significantly associated with an increased risk of stroke. Specifically, the HR for these parameters were 2.61 (95% CI 1.87–3.64), 1.82 (95% CI 1.24–2.67), and 1.01 (95% CI 1.00–1.01), respectively, indicating a strong positive

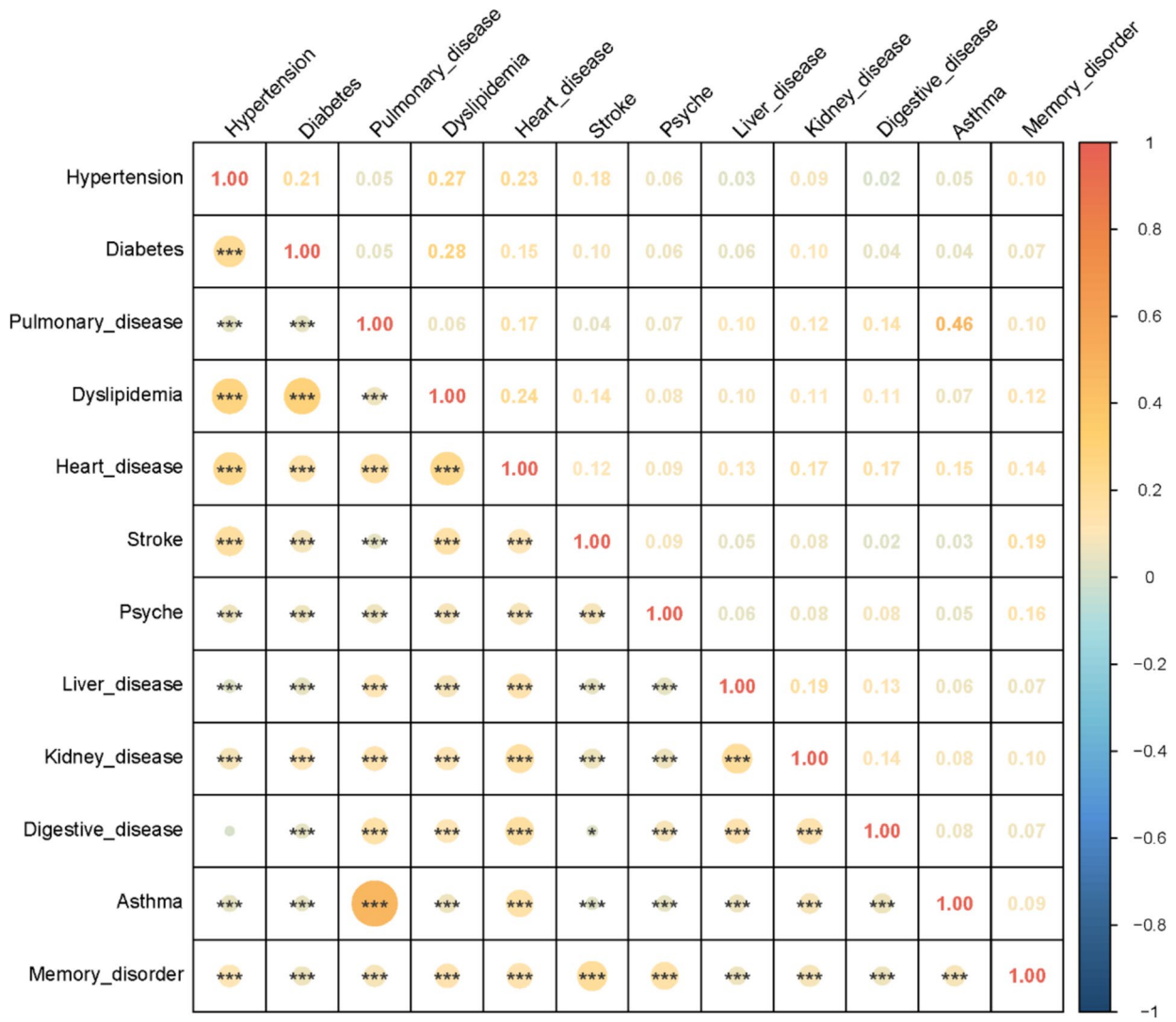


Fig. 1 Correlation matrix of chronic diseases: phi coefficients and statistical significance

correlation between these biomarkers and increased stroke risk. In contrast, higher levels of HDL-C and TMO were significantly associated with a reduced risk of stroke, with HRs of 0.99 (95% CI 0.98–0.99), 1.00 (95% CI 0.99–1.00), respectively, implying that these factors may have a protective effect and help reduce the risk of stroke. During the stepwise adjustment of the model, we noticed that the correlation of WHtR, TyG-WC, TyG-BMI, and TyG with stroke risk changed from a significant correlation to a non-significant correlation. In Model 3, the HRs of these above parameters were 2.05 (95% CI 0.74–5.69), 1.00 (95% CI 0.99–1.00), 1.00 (95% CI 0.99–1.00), and 1.12 (1.00–1.26), respectively. Thus, multiple factors need to be taken into account when assessing stroke risk, and highlights the need for an individualised and multifactorial approach to the assessment and management of stroke risk in clinical practice.

The solid red lines correspond to the central estimates, the gray-shaded regions indicate the 95% confidence intervals, and the blue area represents the density distribution of the predictor.

Development and validation of stroke risk prediction model

Based on the clinical data of stroke patients, we incorporated sex, age, and the nine feature parameters identified by the machine learning algorithms (TyG-WC, WHtR, TyG-BMI, TyG, TMO, CysC, CREA, SBP, HDL-C) as predictive variables to create a nomogram model for predicting the probability of stroke over 1, 3, and 5 years. By summing the scores of these predictive variables, the nomogram model calculates a total score for each patient, which is then used to predict their stroke risk.

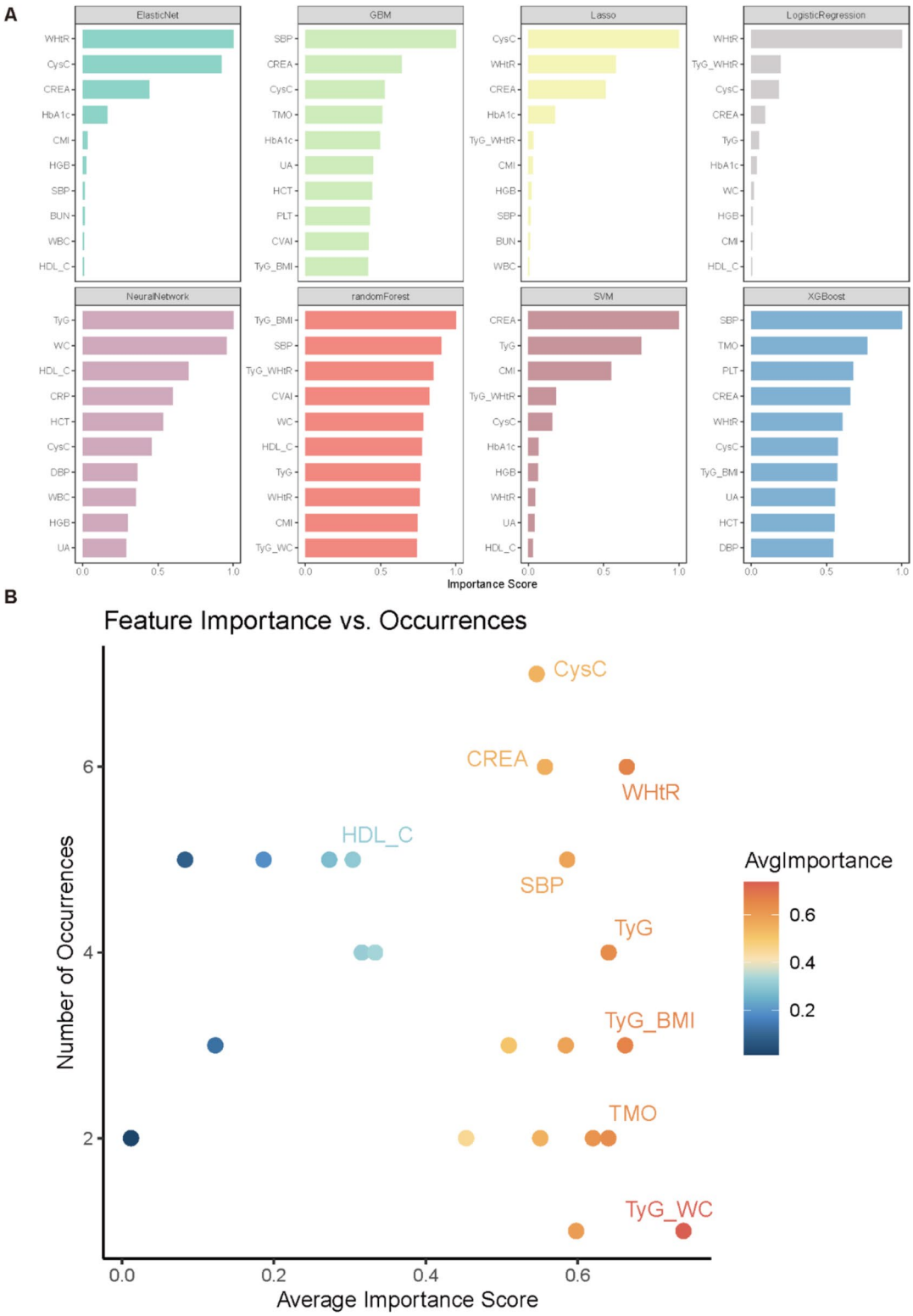


Fig. 2 Feature importance analysis across multiple machine learning algorithms. **A** Bar plots showing the top 10 features and their importance scores for each of the eight machine learning algorithms. **B** Scatter plot of feature importance consensus. Each point represents a feature, with its average importance score on the x-axis and the number of times it appeared in the top 10 features across all algorithms on the y-axis. The top five features by occurrence count and the top five by average importance score are labeled.

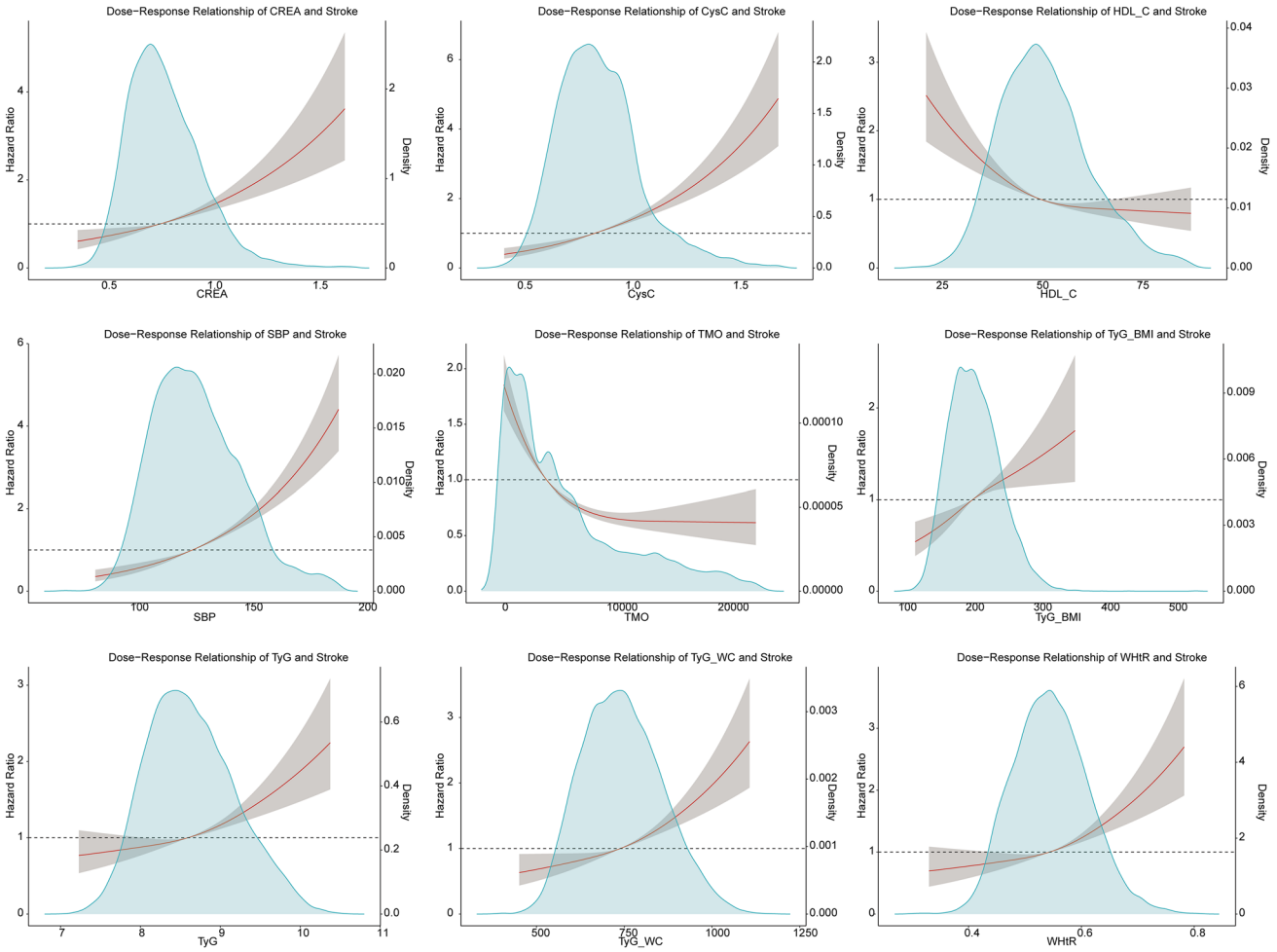


Fig. 3 Dose–response relationships between key predictors and stroke risk

For example, for a 55-year-old male patient, as shown in Fig. 4A, the total score obtained by summing the scores of each predictive factor is 487 points. This corresponds to a stroke probability of 0.556, 0.735, and 1.15% over 1, 3, and 5 years, respectively. The calibration curve for stroke risk prediction is shown in Fig. 4B. The calibration curves demonstrated strong alignment with the 45-degree line, particularly for the 1-year and 3-year predictions, indicating that the predicted probabilities closely matched the actual observed outcomes. This close alignment highlights the reliability and accuracy of our model across different time points. Combined with high AUC values, these results underscore the model's robust performance in both discrimination and calibration, supporting its potential clinical utility for individualized stroke risk assessment and management.

This curve demonstrates that the predicted probabilities for 1, 3, and 5 years are closely aligned with the actual observed probabilities, falling near the 45° line. This indicates that our prediction model has high reliability and accuracyThe Harrell's C index was carefully calculated for time-dependence over the 8-year follow-up period to assess the predictive performance of these characteristic parameters and the nomogram score for stroke risk. The results are shown in Fig. 4C. The nomogram score performed most prominently, with the Harrell's C index remaining above 0.70 over the 1 to 8-year follow-up period, with AUC values ranging from 0.73 (95% CI 0.70–0.77), 0.73 (95% CI 0.69–0.76), 0.71 (95%CI 0.68–0.74), 0.72 (95% CI 0.70–0.75), 0.72 (95% CI 0.69–0.74), 0.71 (95% CI 0.68–0.73), 0.71 (95% CI 0.69–0.73), and 0.71 (95% CI 0.69–0.73), respectively. This result highlights the superiority of the combined assessment compared with the performance of individual predictors, further confirming the reliability and validity of the column-line graph prediction model in predicting stroke risk.

Table 2 HR for stroke risk associated with key predictors across three adjustment models

Index	HR (95% CI)	p value
CysC_Model1	3.82 (2.74–5.32)	< 0.001
CysC_Model2	3.69 (2.65–5.15)	< 0.001
CysC_Model3	2.61 (1.87–3.64)	< 0.001
WHtR_Model1	55.91 (21.55–145.01)	< 0.001
WHtR_Model2	57.93 (22.22–151.06)	< 0.001
WHtR_Model3	2.05 (0.74–5.69)	0.168
CREA_Model1	2.50 (1.72–3.64)	< 0.001
CREA_Model2	2.47 (1.69–3.60)	< 0.001
CREA_Model3	1.82 (1.24–2.67)	0.002
SBP_Model1	1.02 (1.017–1.02)	< 0.001
SBP_Model2	1.02 (1.02–1.02)	< 0.001
SBP_Model3	1.01 (1.00–1.01)	< 0.001
HDL_C_Model1	0.98 (0.98–0.99)	< 0.001
HDL_C_Model2	0.98 (0.98–0.99)	< 0.001
HDL_C_Model3	0.99 (0.98–0.99)	< 0.001
TyG_WC_Model1	1.00 (1.00–1.00)	< 0.001
TyG_WC_Model2	1.00 (1.002–1.00)	< 0.001
TyG_WC_Model3	1.00 (0.99–1.00)	0.258
TyG_BMI_Model1	1.01 (1.01–1.01)	< 0.001
TyG_BMI_Model2	1.01 (1.01–1.01)	< 0.001
TyG_BMI_Model3	1.00 (0.99–1.00)	0.595
TyG_Model1	1.51 (1.35–1.69)	< 0.001
TyG_Model2	1.51 (1.35–1.69)	< 0.001
TyG_Model3	1.12 (0.99–1.26)	0.05001
TMO_Model1	0.99994 (0.99992–0.99995)	< 0.001
TMO_Model2	0.99994 (0.99993–0.99995)	< 0.001
TMO_Model3	0.99995 (0.99994–0.99997)	< 0.001

Model 1 was adjusted for age and sex; model 2 was also adjusted for marital status, place of residence, education, smoking status, and alcohol consumption; and model 3 was further adjusted for hypertension, diabetes mellitus, heart disease, dyslipidaemia, kidney disease, systolic blood pressure, and diastolic blood pressure. Hazard ratios (HR), 95% confidence intervals (CI), and p values were calculated for each predictor in all three models

Discussion

Stroke is a leading cause of disability worldwide, and China faces the largest stroke burden globally. Due to the aging population, the persistent high prevalence of risk factors, and poor management, this burden is expected to increase further in China (Wu et al. 2019). Although some reports suggests a decline in age-standardized stroke incidence, the prevalence and incidence of stroke continue to rise among individuals under 70 years of age, especially with signs of early-onset stroke in younger populations (Potter et al. 2022; Sporns et al. 2022; Janas et al. 2023). This trend toward younger stroke patients adds to the societal burden. Among Chinese adults experiencing their first ischemic stroke, the 28-day mortality rate is reported to be 3%, while 41% of survivors experience recurrent strokes within five years (Chen et al. 2020). Therefore, rapid identification of stroke risk factors is crucial for early screening, timely intervention, and improving treatment, preventing recurrence, and enhancing outcomes. Our study closely examined the risk factors for stroke and found that stroke patients are influenced by multiple factors, including sociodemographic characteristics and comorbidities. We utilized machine learning algorithms to identify nine key parameters associated with stroke, including TyG-WC, CysC, and others. Except for TMO and HDL-C, the remaining parameters were positively associated with stroke risk in a dose-dependent manner. Ultimately, we developed a nomogram model that incorporated these nine parameters and clinical features to predict the 1-, 3-, and 5-year stroke risk. This model provides a valuable tool for early identification of high-risk patients, offering personalized risk assessments and recommendations for interventions.

The acceleration of population aging and urbanization has led to the proliferation of unhealthy lifestyles, increasing exposure to stroke risk factors (Tu and Wang 2021). Aging is characterized by decreased stress tolerance, worsened homeostatic imbalance, and an increased risk of age-related diseases, with the risk of stroke doubling every 10 years after age 55 (Surugiu et al. 2023). These pathological changes are often exacerbated by age-related diseases, further increasing stroke risk (Gallizioli et al. 2023). Gender is also an important, non-modifiable factor. Numerous studies have reported that women have a higher risk of stroke, which may be due to longer life expectancy, pregnancy-related factors, postpartum depression, and possible survival disadvantages (Barthels and Das 2020; Thomas et al. 2021). Our findings reflect these trends, as stroke patients in our study were older, and the proportion of female patients was higher. Many diseases have been widely shown to influence stroke morbidity and mortality (Alloubani et al. 2021; Vasconcelos et al. 2021; Nannoni et al. 2021). Age-related indicators exhibit abnormalities in stroke patients, and while other diseases may show higher prevalence, they may not always exhibit strong correlations with stroke. These outcomes are reflected in our underlying characteristics. These complex health challenges are intertwined and collectively influence stroke risk, prognosis, and overall management strategies. Therefore, it is important to fully consider the impact of various idiosyncratic factors when developing prevention and treatment strategies to achieve more precise medical interventions.

Bonkhoff and Grefkes (2022) suggested that artificial intelligence offers substantial advantages in achieving precision medicine and personalized outcome prediction for stroke. Machine learning has become an effective tool for risk prediction and clinical assessment across various

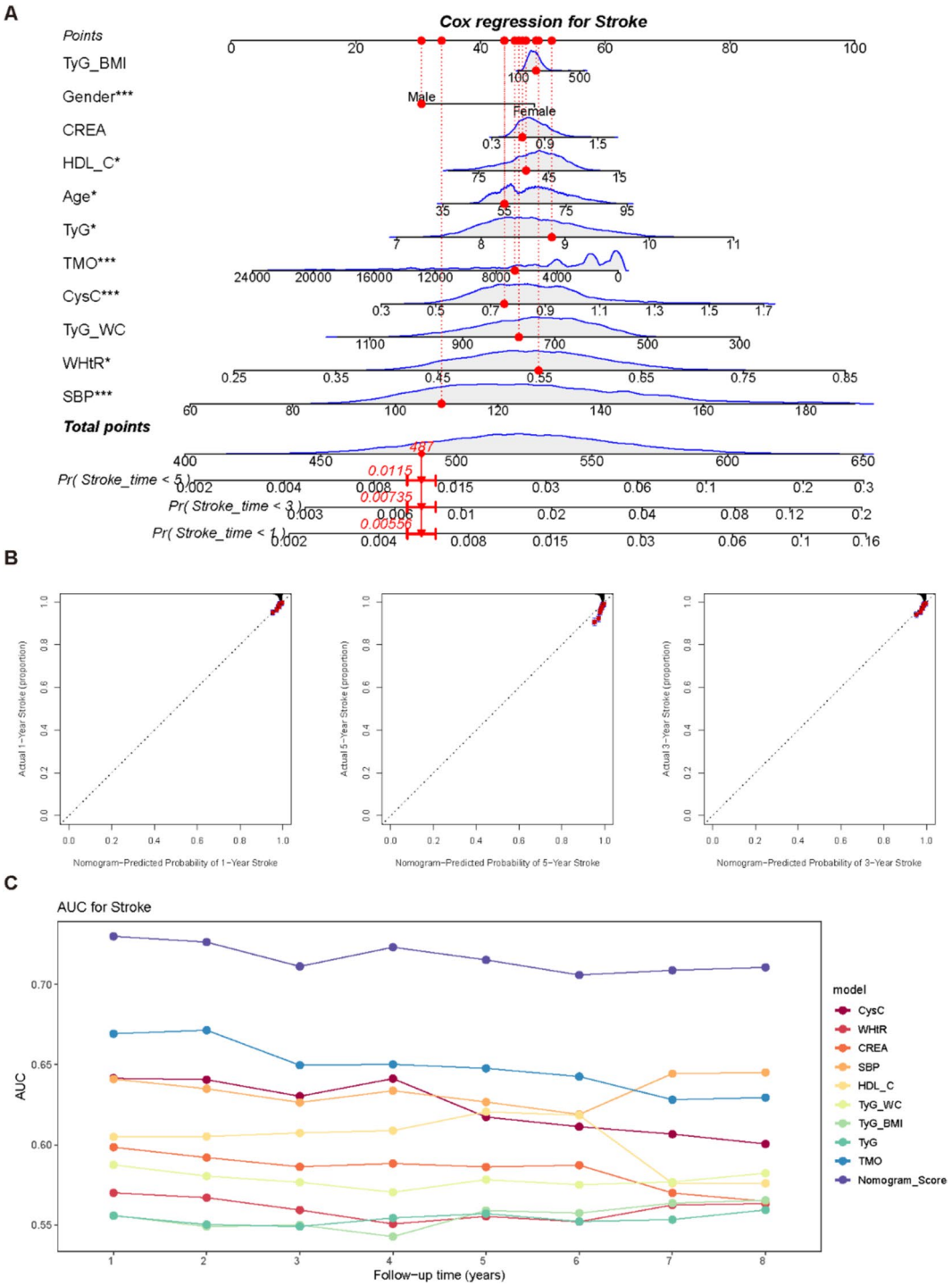


Fig. 4 Development and validation of a stroke risk prediction model. **A** Nomogram for predicting 1-, 3-, and 5-year stroke risk. Points are assigned for each risk factor by drawing a line upward from the corresponding value to the 'Points' line. The sum of these points plotted on the 'Total Points' line corresponds to the predicted 1-, 3-, and 5-year stroke risk. **B** Calibration curves for 1-, 3-, and 5-year stroke risk predictions. The x-axis represents the predicted probability, and the y-axis represents the actual observed probability. Perfect predictions should fall on the diagonal line. Closer alignment of the red line to the diagonal indicates better calibration. **C** Time-dependent AUC curves for individual predictors and the nomogram score over 8 years of follow-up. Higher AUC values indicate better predictive performance. The nomogram score represents the combined predictive power of all included variables.

diseases (Hohmann 2022). Machine learning is increasingly used for diagnosing and predicting stroke and its complications or sequelae, demonstrating robust and accurate results (Schwartz et al. 2023; Sheth et al. 2023; Bidoki et al. 2023). We employed eight machine learning algorithms to identify stroke risk factors and, by combining average importance scores and frequency of occurrence, ultimately identified nine key predictive parameters: TyG-WC, WHtR, TyG-BMI, TyG, TMO, CysC, CREA, SBP, and HDL-C. Huang et al. (2023) found that higher levels of TyG and its related indices (TyG-BMI, TyG-WC) are associated with increased stroke risk, making them reliable indicators for identifying high-risk individuals. Various obesity-related indicators, such as body weight, BMI, WC, and WHtR, have been shown to be significantly associated with stroke, with individuals in higher ranges showing the greatest stroke risk (Lin et al. 2023). TMO is an indicator used to measure metabolic rate, expressed in metabolic equivalents. Park et al. (2023) found that when exercise capacity is low, as indicated by a lower TMO, the risk of cardiovascular disease is higher. Serum CREA is the most widely used biomarker for estimating kidney function. When used in combination with CysC, it enhances assessment accuracy, and recent studies have shown that their combined detection improves stroke risk stratification (Lees et al. 2023). Yim et al. (2023) found a weaker correlation between CREA levels and stroke but identified a positive association in women. A study in Japanese women with cardioembolic stroke also reported that lower creatinine clearance worsened functional outcomes (Sakai et al. 2023). Elevated CysC levels may be an independent risk factor for ischemic stroke and post-stroke cognitive impairment (Yan et al. 2022). BP is considered a major risk factor for stroke mortality (Cao et al. 2021). A study found that for individuals over 50, each 4 mmHg increase in average SBP was associated with a 1% increase in the 5- and 10-year stroke risk (Sethi et al. 2021). Previous epidemiological studies have shown that HDL-C concentration is negatively correlated with cardiovascular disease risk and can be used for risk prediction (Franczyk et al. 2021).

Our dose–response analysis revealed significant positive dose–response relationships between CREA, CysC, SBP, TyG-BMI, TyG, TyG-WC, WHtR, and stroke risk, consistent with previous studies. In contrast, TMO and HDL-C exhibited inverse relationships. Studies have reported that during ischemic stroke, energy metabolism failure leads to mitochondrial dysfunction and oxidative stress, which disrupt the blood–brain barrier and exacerbate brain injury (Zhao et al. 2021; An et al. 2021). Al Rifai et al. (2020) suggested that higher metabolic equivalents are associated with a lower probability of stroke, indicating that higher TMO may reduce stroke risk. Meanwhile, higher HDL-C levels do not always prevent cardiovascular disease. Hu et al. (2023) found that HDL-C is negatively correlated with stroke when its level is below 1.55 mmol/L; however, above 1.55 mmol/L, stroke incidence is not significantly reduced. There is also evidence suggesting a U-shaped relationship between HDL-C levels and the risk of stroke and other cardiovascular diseases (Liu and Cheng 2024). Interestingly, the attenuated statistical significance of some predictors, including TyG-WC, WHtR, and TyG-BMI, highlights potential challenges in translating machine learning findings into traditional epidemiological models. This phenomenon may be attributed to multicollinearity among predictors or mediation effects within complex metabolic pathways. Machine learning models, however, excel in capturing non-linear relationships and interactions, preserving the predictive value of these indices despite their diminished direct associations in adjusted regression models (Kyriazos and Poga 2024). While these variables lose statistical significance in the fully adjusted model, it's important to note that this doesn't necessarily negate their clinical relevance. Future research should focus on disentangling these relationships through advanced modeling techniques and prospective validation to further clarify their role in stroke risk stratification.

Our study also highlights the potential for machine learning to enhance traditional clinical prediction models by incorporating novel biomarkers and indices. Compared to conventional regression approaches, machine learning algorithms offer improved flexibility in handling complex, multi-dimensional data, enabling the identification of subtle, non-linear patterns among predictors (Churpek et al. 2016). These parameters identified by machine learning are not only stroke risk factors but also predictors of stroke risk assessment. However, caution is needed to balance the complexity of such models with their interpretability and clinical applicability. In various cardiovascular diseases, TyG-WC (Huang et al. 2023), WHtR (Lin et al. 2023), TyG-BMI (Huang et al. 2023), TyG (Huang et al. 2023), TMO (Yeo et al. 2021), CysC (Ding et al. 2021), CREA (Chandler et al. 2023), SBP (Yannoutsos et al. 2017), HDL-C (Kim et al. 2022) have demonstrated their predictive capabilities. High-risk patients should have their levels of these parameters regularly monitored, and clinicians ought to adopt more aggressive management strategies when the markers are elevated. The novel indices like TyG-WC and TyG-BMI, although showing promise, demand further validation prior to extensive clinical application. In the meantime, they can serve as supplementary data to improve risk stratification, especially when traditional risk factors cannot fully account for a patient's risk status. Given the importance of metabolic factors, lifestyle modifications focusing on weight management and glucose control should be emphasized. The wide variety of identified risk factors implies the necessity of personalized prevention strategies. Patients with elevated novel risk factors may require closer surveillance and earlier intervention. Future research should focus on prospective studies

to confirm the clinical utility of novel indices and explore targeted interventions based on specific risk factor profiles. By taking these aspects into consideration, clinicians can start applying our findings in practice, while being aware of the continuous need for validating and refining novel risk factors.

In recent years, nomograms have been widely used as tools for risk and prognosis prediction. Compared to traditional staging systems, nomograms have shown superior ability to identify high-risk populations (Wu et al. 2020). Several studies have validated the effectiveness of nomogram models in predicting stroke risk (Wu et al. 2023; Li and Li 2022; Mao et al. 2024). In this study, we developed a comprehensive nomogram model integrating age, gender, and nine key predictors identified using machine learning algorithms, which demonstrated excellent calibration and discrimination for predicting 1-, 3-, and 5-year stroke risk. The C-index, a critical metric for survival analysis and time-to-event data, measures the model's discriminative ability by evaluating its capacity to rank patients according to stroke risk (Longato et al. 2020). With values ranging from 0.5 (random discrimination) to 1 (perfect discrimination) (Che Nawi et al. 2023), our model consistently achieved a C-index above 0.70 over the 8-year follow-up period, demonstrating substantial discriminative power. This robustness is particularly valuable in clinical settings, where accurate risk stratification is essential for guiding preventive and therapeutic interventions. The AUC, is the area of the Receiver Operating Characteristic, widely used in binary outcome prediction to assesses the model's ability in distinguishing disease risk between individuals (Dritsas and Trigka 2022). When the AUC exceeds 0.7, it indicates a good predictive performance of the model (Mandrekar 2010). Our results objectively demonstrate the robustness and reliability of our prediction model in predicting stroke risk. In addition, the predictions of a well-calibrated model should closely match observations over the entire range of predicted probabilities (Steyerberg et al. 2010). Our calibration curves fit the 45-degree line, suggesting that the predicted probability of stroke is similar to the actual observed probability, demonstrating the reliable risk estimates of our model at different levels of risk. Together, these metrics affirm the model's clinical utility and its potential to support effective decision-making in stroke prevention and management.

The clinical utility of our study lies in providing a user-friendly and individualized risk assessment tool for early stroke detection, risk stratification, and targeted preventive interventions. To facilitate its implementation in clinical practice, several steps are needed: (1) external validation in diverse populations to ensure generalizability; (2) prospective studies evaluating its impact on clinical decision-making and patient outcomes; (3) integration into electronic health records for seamless use in real-world settings; (4) development of clinical guidelines for interpreting nomogram scores and translating them into management strategies; and (5) education and training for healthcare providers on proper use of the tool. Further research should focus on refining the model by incorporating emerging biomarkers or genetic factors, assessing cost-effectiveness in routine care, and evaluating its performance in specific subgroups, such as varying age groups and comorbidity profiles. With these advancements, the nomogram holds significant potential as a practical and impactful tool for improving stroke prevention and clinical outcomes.

Conclusion

The study reveals that stroke is influenced by multiple factors. Using machine learning algorithms, we successfully identified nine parameters significantly associated with stroke risk, including TyG-WC, WHtR, TyG-BMI, TyG, TMO, CysC, CREA, SBP, and HDL-C. Among these, TyG-WC, WHtR, TyG-BMI, TyG, CysC, CREA, and SBP showed a positive dose–response relationship with stroke risk, while TMO and HDL-C exhibited an inverse relationship. Furthermore, we developed a nomogram prediction model incorporating these parameters and clinical data to estimate the 1-, 3-, and 5-year stroke risk for patients. This tool demonstrates great potential in the early identification of high-risk individuals and the formulation of personalized intervention strategies.

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Data availability This study exclusively utilized publicly available data, and the sources and handling of these data are thoroughly described in the Materials and Methods section. Additional information can be obtained from the corresponding author upon request.

Declarations

Conflict of interest The authors declare no conflict of interest.

Informed consent Not applicable.

Institutional review board Not applicable.

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